

Authority physical page 1 — provenance cover; excluded copy matter.

Identity evidence retained: the vector NUMDAM cover names Pierre Deligne and Jean-Pierre Serre jointly, gives the title “Formes modulaires de poids 1,” and supplies the publication locator Annales scientifiques de l’É.N.S., 4e série, tome 7, no 4 (1974), pp. 507–530. It also contains the NUMDAM item locator, copyright notice, terms-of-use notice, and digitization-program statement.

Editorial disposition: EXCLUDE_COPY_MATTER_RETAIN_PROVENANCE. The French and English body records are explicitly empty and complete. None of the cover’s legal, copyright, access, or digitization boilerplate is edition content. The title, both authors, and publication identity are retained only as metadata and apparatus evidence.

Comparator: none; the 24-page comparator begins with authority physical page 2.

Salvage: no candidate member supplies an authority-faithful page-local cover record; all salvage remains ZERO_ACCEPTED.

Assets: none. The cover is not ingested as edition content and no crop fallback is required.

Unresolved readings: none.

Authority physical page 2 / article page 1 / printed page 507.

Included matter: article title, exact joint byline, Cartan dedication, Introduction, the explicit §§ 1–9 plan, and the beginning of § 1 through the subgroup chain $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N)$. Journal locator matter above the title and the journal footer below the text are not part of the reading body.

French replay decisions: “A Henri Cartan” is retained as visibly printed; the mathematical symbols $\overline{Q}$, $GL_2(\mathbb{C})$, the prime ℓ , section signs, guillemets, and punctuation were checked against the authority render. The page was semantically rendered and reviewed before its French hash was frozen.

Comparator: comparator page 1. Direct rendered review found the same page and geometry; normalized extracted text is identical (similarity 1.000000). The comparator remains lower-raster and noncontrolling.

Salvage: the partial candidate contains the opening matter but alters apostrophes, quotation marks, dedication typography, and page geometry. No candidate bytes or readings were accepted.

English: translated only from the frozen French record; title, two-author byline, dedication, section plan, claims, and numbering are preserved.

Assets: none; semantic text and mathematics suffice. Unresolved readings: none.

Authority physical page 3 / article page 2 / printed page 508.

Included matter: completion of 1.1 and §§ 1.2–1.5. High-risk checks covered the three matrix membership equivalences for $\Gamma(N)$, $\Gamma_1(N)$, and $\Gamma_0(N)$; the slash operator $(f|_k \gamma)(z)$; the Möbius action γz ; conditions (1.2.1)–(1.2.3); $e^{2\pi iz/N}$; the Fourier expansion and $f_\infty(q)$; R_d ; and the parity condition for ε .

Running head, printed folio, and journal footer were omitted from the edition body. Line-end hyphenation was resolved semantically without changing wording.

Comparator: comparator page 2; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate largely follows the analytical definitions but normalizes notation and typography and lacks page-local pixel proof. It remains ZERO_ACCEPTED.

English: standalone translation from the frozen French; “holomorphe aux pointes” is rendered “holomorphic at the cusps,” and “forme modulaire parabolique” as “cusp form,” with all formulas and numbering unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 4 / article page 3 / printed page 509.

Included matter: continuation of 1.5, §§ 1.6–1.8 through the first case of the local rule. Formula checks covered $f \mid R_d = \varepsilon(d)f$, the transformation law with $\varepsilon(d)$, the quotient map to $d \pmod{N}$, Hecke formulas (1.6.1) and (1.6.2), the normalized newform expansion, Dirichlet series (1.7.1), Euler product (1.7.2), and the first $p \mid N$ case in 1.8.

Source-faithful decision: the printed transformation law uses $\varepsilon(d)$. The salvage candidate's $\varepsilon(a)^{-1}$ is mathematically related on $\Gamma_0(N)$ but is not the source reading and was rejected. Formula numbering and source conventions for unbounded summation signs were preserved.

Comparator: comparator page 3; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: besides the altered character formula, the candidate rewrites the quotient-map presentation and normalizes typography. All candidate text remains ZERO_ACCEPTED.

English: translated from the frozen French with “primitive cusp forms (‘newforms’)” and the complete Euler product retained.

Assets: none. Unresolved readings: none.

Authority physical page 5 / article page 4 / printed page 510.

Included matter: completion of 1.8, § 1.9, the start of § 2, §§ 2.1–2.2 through equation (2.2.1). High-risk checks covered the exponents $(k-1)/2$ and $k/2-1$; the decomposition $f(z) = E(z) + \sum \lambda_i f_i(d_i z)$; μ_N ; $\omega_E^{\otimes k}$; $\omega_E = e^* \Omega_{E/S}^1$; the labels (a), (b), (c); generalized elliptic curves; $\tilde{f}$; and $(f | R_d)(E, \alpha) = f(E, d\alpha)$.

Source-faithful decisions: λ_i is retained (the candidate substitutes ψ_i); the source's Roman labels (a), (b), (c) are retained (the candidate substitutes Greek labels). Running matter and folio were excluded.

Comparator: comparator page 4; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the lambda/psi substitution, label substitutions, and degraded punctuation are material divergences. No salvage reading was accepted.

English: translated only from the frozen French; “parabolique” is rendered by the standard “cusp form” terminology without changing the geometric vanishing condition.

Assets: none. Unresolved readings: none.

Authority physical page 6 / article page 5 / printed page 511.

Included matter: continuation of 2.2 and §§ 2.3–2.5 through equation (2.5.1).

Formula checks covered the geometric Hecke action

$$(f \mid T_p)(E, \alpha) = p^{-1} \sum_{\varphi} \varphi^* (f(\varphi E, \varphi \circ \alpha));$$

$\alpha(\exp(2\pi i/N))$; (2.4.1) with $1/N$ and $(2\pi i du)^{\otimes k}$; (2.4.2) with Id and $(dt/t)^{\otimes k}$; the

Tate curve $\mathbf{G}_m/q^{\mathbb{Z}}$; $a_n(d)$; and (2.5.1) with $a_n(p)q^{np}$.

Source-faithful decisions: the authority's $p^{-1} \sum \varphi^*$ formula, $1/N$, du , Id , $a_n(d)$, and $a_n(p)$ were retained. The candidate changes each of these and was rejected. The journal footer and printed folio were omitted.

Comparator: comparator page 5; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: substantial mathematical substitutions in §§ 2.3–2.5 make the candidate unusable as authority text; all related PDF derivatives inherit the errors. Accepted salvage members remain zero.

English: standalone translation from the frozen French; all normalization factors, pullbacks, tensor powers, coefficient notation, and equation numbers are unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 7 / article page 6 / printed page 512.

Included matter: equation (2.5.2); § 2.6; Proposition 2.7 and its proof; Remark 2.8; the § 3 heading; and the beginning of 3.1 through $\rho : G \rightarrow \mathrm{GL}_n(k)$. High-risk checks covered the two sums in (2.5.2), $S_K = K \otimes_{\mathbf{Q}} S_{\mathbf{Q}}$, the defining lattice condition $(f \mid R_d)_{\infty}(q) \in \mathbf{Z}[[q]]$, assertions (2.7.1)-(2.7.4), bounded denominators, $\mathbf{Z}((q)) \otimes \mathbf{Q}$, and the multiplication by Δ .

Source-faithful decision: Proposition 2.7 requires integral q -series in $\mathbf{Z}[[q]]$; the salvage candidate changes this to $\mathbf{Z}[[q]] \otimes \mathbf{Q}$, which is not the source reading. The full proof, Remark 2.8, § 3 heading, and 3.1 were retained; the candidate omits them. Running head, folio, and journal footer were excluded.

Comparator: comparator page 6; direct review found the same page and geometry, with normalized text identity 1.000000. The comparator remains lower-raster and noncontrolling.

Salvage: the candidate is structurally incomplete at the § 2/§ 3 boundary and changes the defining integrality condition. All candidate members remain ZERO_ACCEPTED.

English: standalone translation from the frozen French; “champ algébrique” is rendered “algebraic stack,” and “formes modulaires paraboliques” as “cusp forms,” with formulas and numbering unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 8 / article page 7 / printed page 513.

Included matter: continuation of 3.1; the three permitted coefficient-field types; the definition of being unramified at p ; $F_{p,\rho}$ and polynomial (3.1.1); Lemma 3.2 and its proof; Remarks 3.3-3.4; the § 4 heading; subsection (a); the beginning of Theorem 4.1; and footnote (1). Formula checks covered $P_{p,\rho}(T) = \det(1 - F_{p,\rho}T)$, its trace-determinant expansion, the characteristic-zero trace alternative, and the conditions $\varepsilon(-1) = -1$ and $p \nmid N$.

Source-faithful decisions: the coefficient field is denoted k ; the Frobenius element is $F_{p,\rho}$; and footnote (1) identifies the source convention with the element φ in [5]. The salvage candidate omits nearly all of § 3 on this page and the footnote, then jumps to Remarks 3.3-3.4 and § 4.

Comparator: comparator page 7; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the missing coefficient-field cases, Frobenius definition, characteristic polynomial, Lemma 3.2, proof, and footnote prevent any page-local reuse. Accepted salvage remains zero.

English: translated only from the frozen French; “substitution de Frobenius” is retained as “Frobenius substitution,” and the unramified/semisimple terminology and footnote are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 9 / article page 8 / printed page 514.

Included matter: completion of Theorem 4.1 and equation (4.1.1); Corollary 4.2; Remarks 4.3-4.5; subsection (b); Theorem 4.6; and Corollary 4.7. High-risk checks covered $\rho : G \rightarrow \mathrm{GL}_2(\mathbf{C})$, $\mathrm{Tr}(F_{p,\rho}) = a_p$, $\det(F_{p,\rho}) = \varepsilon(p)$, $\det \rho = \varepsilon$, the complex-conjugation matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, the Artin conductor, and $\Phi_f(s) = \sum_{n \geq 1} a_n n^{-s}$.

Source-faithful decisions: the page consistently uses $F_{p,\rho}$ rather than a generic Frobenius macro, retains the exact irreducibility/cuspidality equivalence, and keeps the distinction between the representation, its determinant character, and the Artin L -function. Running matter was omitted.

Comparator: comparator page 8; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate normalizes $F_{p,\rho}$ to a generic Frob_p notation and suppresses source-level context and page typography. No candidate reading was accepted.

English: translated from the frozen French; “parabolique” is rendered “cusp form,” while the theorem, conductor, local-factor terminology, and all formulas remain unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 10 / article page 9 / printed page 515.

Included matter: Corollary 4.8 and the first three stages of the proof of Theorem 4.6 through equation (4.9.1) of Lemma 4.9. High-risk checks covered $\tilde{f} = \sum \overline{a_n} q^n; f(-1/Nz) = \lambda z \tilde{f}(z); \mu = i\lambda/N^{1/2}; \Psi_f, \tilde{\Psi}_f, \tilde{\zeta}(s, \rho)$, and the conjugate representation $\bar{\rho}; F(1-s) = \omega \tilde{F}(s); A = (N/M)^{1/2}$; the quotient defining $F_p(s)$; and the finite Euler products in Lemma 4.9.

Source-faithful decisions: the tilde and complex conjugations are substantive, not decorative. The authority's $\Psi_f/\tilde{\Psi}_f$ and $\tilde{\zeta}(s, \rho)/\tilde{\zeta}(s, \bar{\rho})$ equations, the factor i in μ , and the numerator/denominator orientation of $F_p(s)$ were retained exactly.

Comparator: comparator page 9; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate replaces $\tilde{f}$ by f , changes the functional equations and constants, loses the conjugate representation, and introduces different $\tilde{\zeta}/\zeta$ notation. These are material mathematical substitutions; all salvage remains ZERO_ACCEPTED.

English: standalone translation from the frozen French; functional-equation notation, conjugations, Euler factors, and equation numbering are unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 11 / article page 10 / printed page 516.

Included matter: condition (4.9.2), the proof of Lemma 4.9, completion of the proof of 4.6, subsection (c), properties (i)-(iii), and Theorem 4.10. Formula checks covered $(1 - \alpha_p^{(i)} p^{-s})^{\pm 1}$, the zero/pole locations, the exclusion $\alpha_p^{(i)} \neq p/\alpha_p^{(j)}$, Rankin's bound, $f_\chi = \sum \chi(n) a_n q^n$, and the defining data $L(s, \rho), f, \varepsilon = \det \rho$, and $N = \text{conduct. } \rho$.

Source-faithful decisions: the exponent ± 1 and the double indexing of the $\alpha_p^{(i)}$ are retained; the proof compares $\alpha_p^{(i)}$ with $p/\alpha_p^{(j)}$, not merely with $p^{1/2}$. The explicit twisted cusp form f_χ is part of condition (iii) and was preserved.

Comparator: comparator page 10; same page and geometry on direct review; normalized text identity is 0.999327, the small extraction difference being noncontrolling and not visible as a content divergence.

Salvage: the candidate drops the ± 1 exponent and indexed roots, changes the crucial zero/pole exclusion, and omits f_χ . No candidate text was accepted.

English: translated only from the frozen French; the Artin-conjecture condition, twist, Weil-Langlands theorem, and all formulas and numbering are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 12 / article page 11 / printed page 517.

Included matter: completion of the proof of Theorem 4.10; Remarks 4.11-4.14; and the beginning of the examples through the page-boundary split “cor-”. High-risk checks covered the series $\sum a_n \chi(n) n^{-s}$, the T_p and U_p eigenform statement, the primitive form f' on $\Gamma_0(N')$, $f' \mid T_p = a_p f'$, the weakened condition (iii'), the bijection in 4.13, the bound $A \geq (N/12) \prod_{p \mid N} (1 + p^{-1})$, ρ_f , and the projective groups $\mathfrak{A}_4$, $\mathfrak{S}_4$, and $\mathfrak{A}_5$.

Source-faithful decisions: the complete primitivity argument and the qualifications in 4.12-4.14 were retained. The source's physical page ends in the split word “cor-”; this boundary marker is retained in the French page record rather than silently importing text from physical page 13. Running matter was excluded.

Comparator: comparator page 11; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate compresses the proof to a few sentences, omits the f' construction and uniqueness argument, abridges 4.12-4.14, and merges examples from the next source page. It is not page-local authority evidence; accepted salvage remains zero.

English: standalone translation from the frozen French; the argument, qualifications, group notation, and page-boundary continuation are preserved without importing the next page.

Assets: none. Unresolved readings: none.

Authority physical page 13 / article page 12 / printed page 518.

Included matter: completion of the examples following 4.14; the § 5 heading; Proposition 5.1 and (5.1.1); and the first part of Proof 5.2 through the page-boundary phrase “dans tout le.” High-risk checks covered the $N=133$ projective group $\mathfrak{A}_4$; the inequality $\sum_{p \mid N} |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1)$; $\det(\varphi_p) = \varepsilon(p)p^{k-1}$; the tensor product $\varphi_p \otimes \bar{\varphi}_p$; all four conjugated Euler factors; $\lambda_p \bar{\lambda}_p \mu_p \bar{\mu}_p = p^{2k-2}$; and $H(s) = \prod_{p \mid N} (1 - p^{-2s+2k-2})(1 - |a_p|^2 p^{-s})$.

Source-faithful decisions: the page begins with “respond,” completing the split “cor-” on physical page 12, and ends before importing the next page’s “plan complexe.” The authority’s $\mathfrak{A}_4$, inequality sign, complex conjugations, and both factors of $H(s)$ were retained exactly. Running matter was omitted.

Comparator: comparator page 12; same page and geometry on direct review. Normalized text identity is 0.999306, the small extraction difference being noncontrolling and not visible as a content divergence.

Salvage: the candidate changes $\mathfrak{A}_4$ to $\mathfrak{S}_4$, replaces the source inequality by an asymptotic equivalence, drops complex conjugations in the four-factor product, and gives a different $H(s)$. These are material mathematical divergences; accepted salvage remains zero.

English: translated only from the frozen French; “forme parabolique” is rendered “cusp form,” while the page-local continuation, Rankin statement, all conjugations, products, and equation numbering are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 14 / article page 13 / printed page 519.

Included matter: completion of Proof 5.2; Remarks 5.3 and equations (5.3.1)-(5.3.2); 5.4 and definition (5.4.1); and the beginning of Proposition 5.5 through “de nombres.” Formula checks covered $g_m(s) = \sum_{p \mid N} |\mathrm{Tr}(\varphi_p^m)|^2 p^{-ms}/m$, $g(s) = \sum_{m \geq 1} g_m(s)$, the Landau argument, the distinction between the inequality from 5.1 and equality (5.3.1), $\sum_{p \leq x} |a_p|^2 p^{-(k-1)} \sim x$, and the limsup quotient defining $\mathrm{dens. sup} X$.

Source-faithful decisions: the factor $1/m$, exponent p^{-ms} , and weight $p^{-(k-1)}$ in (5.3.2) are substantive. The French record stops at the physical-page boundary “de nombres”; the English correspondingly stops after “of prime,” without importing “numbers” from physical page 15.

Comparator: comparator page 13; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate compresses g_m , omits its full definition and the $1/m$ factor, substitutes asymptotic symbols for source inequalities/equalities, and changes (5.3.2) to a different prime-number asymptotic. No candidate text was accepted.

English: standalone translation from the frozen French; “densité supérieure” is rendered “upper density,” and the analytic statements, limiting directions, and physical-page boundary are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 15 / article page 14 / printed page 520.

Included matter: completion and proof of Proposition 5.5; Remark 5.6; the § 6 heading and subsection (a); and Theorem 6.1 with equation (6.1.1). High-risk checks covered the two conclusions for X_η and Y_η ; $Y(c)$ and $X(c)$; the inequalities involving $\sum_\sigma |\sigma(a_p)|^2$; $r = [K : \mathbf{Q}]$ and $c \geq r/\eta$; the authority's singular heading "Représentation ℓ -adiques"; K_λ ; $\rho_\lambda : G \rightarrow \mathrm{GL}_2(K_\lambda)$; and $\det(F_{\rho_\lambda, p}) = \varepsilon(p)p^{k-1}$ for $p \nmid N\ell$.

Source-faithful decisions: the proof uses upper bounds, not asymptotic equivalences. The visible source heading's grammatical singular/plural mismatch was preserved rather than silently corrected. The Frobenius element remains $F_{\rho_\lambda, p'}$, and the residual prime is represented by ℓ .

Comparator: comparator page 14; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate rewrites the density proof with asymptotic equivalences, regularizes the section heading, and substitutes a generic Frobenius macro in Theorem 6.1. It is not an authority-faithful page record; accepted salvage remains zero.

English: translated only from the frozen French; the source heading is rendered in the singular, and all field, place, ramification, trace, determinant, and density notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 16 / article page 15 / printed page 521.

Included matter: Remark 6.2; Corollary 6.3; Remarks 6.4-6.5; subsection (b); definitions 6.6 and equation (6.6.1); the beginning of Theorem 6.7; and footnote (2). High-risk checks covered the authority wording “6.2 sera redémontré”; ρ_λ , K_λ , O_λ , $\mathfrak{m}_\lambda$, and k_λ ; the distinction between λ -integrality and congruence to zero; $f \mid T_p - a_{pf} \equiv 0 \pmod{\lambda}$; the field k_f ; $\rho : G \rightarrow \mathrm{GL}_2(k_f)$; and the Schur-index citation “[20], IX a.”

Source-faithful decisions: the apparent source cross-reference “6.2 sera redémontré” was retained, not emended to 6.1. The displayed $\rho : G \rightarrow \mathrm{GL}_2(k_f)$ has no terminal full stop because the theorem sentence continues on physical page 17. Footnote (2) belongs to this page and was included in both editions. Running head and journal footer were omitted.

Comparator: comparator page 15; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate silently changes 6.2 to 6.1 and then omits Remarks 6.4-6.5, all of 6.6, the opening of Theorem 6.7, and footnote (2). Most of the page is absent; accepted salvage remains zero.

English: standalone translation from the frozen French; “réalisable sur” is rendered “realizable over,” and the source cross-reference, congruences, valuation-ring notation, and footnote are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 17 / article page 16 / printed page 522.

Included matter: completion of Theorem 6.7 and equation (6.7.1); the proof heading; reductions 6.8-6.10; Lemma 6.11; its proof; and the variant. Formula checks covered $\text{Tr}(F_{\rho,p}) = a_p$ and $\det(F_{\rho,p}) \equiv \varepsilon(p)p^{k-1} \pmod{\lambda}$ in (6.7.1); the tuple $(K', \lambda', f', k', \varepsilon', (a'_p))$; congruences $\text{mod } \lambda'$ and $\text{mod } (\ell-1)$; $E_n \equiv 1 \pmod{\ell}$; $M' = O' \otimes_O M$; $O \cap \mathfrak{m}' = \mathfrak{m}$; $a'_T \equiv a_T \pmod{\mathfrak{m}'}$; the algebra $\mathcal{H}$, character χ , prime ideal $\mathfrak{p}$, map χ' , support argument, and triangular matrices (T_{ij}) .

Source-faithful decisions: the source asserts lifting of eigenvalues, not eigenvectors, and this distinction was preserved. The displayed tensor product, all residue fields and ideals, and the final condition $T_{ii} \equiv a_T \pmod{\mathfrak{m}}$ were retained without modernization.

Comparator: comparator page 16; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: after Corollary 6.3 the candidate jumps to later § 8 material and omits the entirety of this authority page, including Lemma 6.11 and its proof. No page-local source text is recoverable or accepted.

English: translated only from the frozen French; discrete-valuation, residue-field, support, indecomposability, and eigenvalue-lifting terminology is kept precise, with all congruences and numbering unchanged.

Assets: none. Unresolved readings: none.

Authority physical page 18 / article page 17 / printed page 523.

Included matter: 6.12, completing the proof of Theorem 6.7; equation (6.12.1); Lemma 6.13 and its proof; the § 7 heading; and 7.1. High-risk checks covered T_ℓ ; $\rho_\lambda : G \rightarrow \mathrm{GL}_2(K_\lambda)$; $\widehat{O}_\lambda$ and $\tilde{\rho}_\lambda$; semisimplification φ ; $F_{\varphi,p}$; coefficients of $\det(1-sT)$ in k_f ; $\varphi : \Phi \rightarrow \mathrm{GL}_n(k')$; the finite-field Brauer-group and Schur-index argument; $\mathbf{F}_\ell = \mathbf{Z}/\ell\mathbf{Z}$; and property $C(\eta, M)$ with $|H| \geq (1-\eta)|G|$.

Source-faithful decisions: hats, tildes, and the distinction between ρ_λ , $\tilde{\rho}_\lambda$, and its semisimplification φ are substantive and were retained. The source § 7 begins on this physical page and was replayed from authority pixels rather than inferred from the candidate.

Comparator: comparator page 17; same page and geometry on direct review, with normalized text identity 1.000000.

Salvage: this is the known candidate break. The candidate omits 6.12, Lemma 6.13, the § 7 heading, and 7.1, and therefore supplies no acceptable page-local text.

English: standalone translation from the frozen French; “réalisable sur” is rendered “realizable over,” “représentation identique” as “identity representation,” and all finite-field, Frobenius, and cardinality notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 19 / article page 18 / printed page 524.

Included matter: Proposition 7.2 and its complete proof, finishing § 7. High-risk checks covered the hypotheses $\eta < 1/2$ and $M \geq 0$; the four subgroup cases involving $SL_2(\mathbb{F}_\ell)$, Cartan subgroups, Cartan normalizers, and projective images $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$; the cardinality $|G| = r\ell(\ell^2 - 1)$; the counts $\ell^2 + \ell$, ℓ^2 , and $\ell^2 - \ell$ for fixed characteristic polynomial; and the bounds $2M/(1 - \eta)$, $4M/(1 - 2\eta)$, and $120M/(1 - \eta)$.

Source-faithful decisions: the page begins directly with Proposition 7.2, continuing the § 7 framework established on physical page 18. All case labels, projective group symbols, inequality directions, and factors of ℓ were retained; running head, folio, and journal footer were omitted.

Comparator: comparator page 18; same article page and geometry on direct review, with normalized text identity 1.000000. The comparator is lower-raster and noncontrolling.

Salvage: the candidate omits Proposition 7.2 and its proof in full, including all four cases and every quantitative bound. No page-local candidate text was accepted.

English: translated only from the frozen French; “sous-groupe de Cartan” is rendered “Cartan subgroup,” and all classification, cardinality, and bounding notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 20 / article page 19 / printed page 525.

Included matter: the § 8 heading; 8.1 and the Eisenstein-series case; 8.2 and the construction of the residual representations; Lemmas 8.3 and 8.4; and 8.5 through the page-end congruence. High-risk checks covered $\rho = \chi_1 \oplus \chi_2; O_K$; primes ℓ splitting completely in K ; places $\lambda_\ell; \rho_\ell : \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \rightarrow \text{GL}_2(\mathbf{F}_\ell)$; $\det(1 - F_{\rho_\ell, p} T) \equiv 1 - a_p T + \varepsilon(p) T^2 \pmod{\lambda_\ell}$; the image $G_\ell; X_\eta, M, H_\ell$; the Čebotarev bound $|H_\ell| \geq (1 - \eta)|G_\ell|$; and the finite set Y of products $(1 - \alpha T)(1 - \beta T)$.

Source-faithful decisions: representation-first Frobenius subscripts, the indexed place λ_ℓ , and the distinction between M and its cardinality M are substantive and were preserved. The French page ends with the congruence modulo λ_ℓ and does not import the finite- Y conclusion from the next page.

Comparator: comparator page 19; same article page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate omits the § 8 heading, 8.1, 8.2, Lemmas 8.3-8.4, and 8.5. It contains none of the page's bounded-image argument, indexed residual representations, or page-end congruence; accepted salvage remains zero.

English: standalone translation from the frozen French; “se décomposent complètement” is rendered “split completely,” “corps résiduel” as “residue field,” and all ramification, determinant, density, and finite-set notation is retained.

Assets: none. Unresolved readings: none.

Authority physical page 21 / article page 20 / printed page 526.

Included matter: completion of 8.5; all of 8.6; and 8.7 through the two $O(1)$ character sums at the physical-page boundary. High-risk checks covered the finite- Y equality; the definition of L' and the condition $R \not\equiv S \pmod{\lambda_\ell}$; lifting the identity representation from $\mathbf{F}_\ell$ to O_{λ_ℓ} ; $\rho : G \rightarrow \mathrm{GL}_2(O_{\lambda_\ell})$; roots of unity of order $\leq A$; the determinant congruence and equality for $F_{\rho,p}$; comparison with ρ' over $\lambda_{\ell'}$; and the exact character-sum decomposition involving $\chi_1\overline{\chi_2}$ and $\chi_2\overline{\chi_1}$.

Source-faithful decisions: $F_{\rho,p}$ and $F_{\rho',p}$ retain representation-first subscripts; λ_ℓ and $\lambda_{\ell'}$ remain distinct; and the proof uses two bounded cross-character sums rather than an asserted asymptotic equivalence. The record stops before the next page's "On en tire."

Comparator: comparator page 20; same page and geometry on direct review. Normalized text identity is 0.999466, with the small extraction difference noncontrolling and no visible content divergence.

Salvage: the candidate gives only a compressed, displaced paraphrase of parts of 8.6-8.7, uses generic Frobenius and place notation, omits the two-lift comparison in full, and replaces the exact character-sum argument by an unsupported asymptotic statement. No candidate text was accepted.

English: translated only from the frozen French; "anneau de la valuation" is rendered "valuation ring," and all lifting, congruence, Frobenius, irreducibility, and character-conjugation notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 22 / article page 21 / printed page 527.

Included matter: the concluding contradiction for § 8; the § 9 heading; the Fourier expansion for a weight-one form; subsection (a); Theorem 9.1; Corollary 9.2; and the proof of 9.1 through the two local cases and displayed estimate (9.3). High-risk checks covered $\sum |a_p|^2 p^{-s} = 2 \log(1/(s-1)) + O(1)$; $f = \sum_{n=0}^{\infty} a_n e^{2\pi i n z/M}$; $|a_n| = O(d(n))$; $d_N(n)$; the labels (ii₁) and (ii₂); $a_n = (a_p)^m$ when $p \mid N$; and $a_n = \lambda^m + \lambda^{m-1}\mu + \dots + \lambda\mu^{m-1} + \mu^m$ when $p \nmid N$.

Source-faithful decisions: the authority's exponential Fourier variable was retained instead of normalized to q ; the ramified case states $|a_n| \leq 1$, not equality; and the displayed label (9.3), case labels, and page endpoint were preserved. Running matter was omitted.

Comparator: comparator page 21; same article page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate shortens the page, changes the Fourier expansion to $\sum a_n q^n$, compresses the proof, omits the displayed and case structure, and incorrectly asserts equality $|a_{p^m}| = 1$ in the ramified case. These are material divergences; accepted salvage remains zero.

English: standalone translation from the frozen French; "premiers à N " is rendered "prime to N ," and all divisor functions, local cases, powers, and theorem numbering are preserved.

Assets: none. Unresolved readings: none.

Authority physical page 23 / article page 22 / printed page 528.

Included matter: Remark 9.4; subsection (b); Proposition 9.5; Lemma 9.6 and its proof; and the first part of the proof of 9.5 through $\log |a_{n(x)}| \sim i(x) \log 2$. High-risk checks covered $|b_n| = O(n^{(k-1)/2} d(n))$; the maximal-order expression $2^{\log n / \log \log n}$; both limsup formulas; the positive-density sets X_N and Y_N ; the operators $g \mid T_p$ and $g \mid R_p$; the matrix identity $F_{\rho_i, p} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; conjugacy to $\rho_i(c)$; and $a_{n(x)} = 2^{i(x)} a_m$.

Source-faithful decisions: the citation “[5], 8.2,” the hypothesis $M \geq 1$, the congruences $p \equiv \pm 1 \pmod{N}$, and the Čebotarev construction were retained. The record ends after the logarithmic asymptotic and does not import the density calculation from physical page 24.

Comparator: comparator page 22; same page and geometry on direct review. Normalized text identity is 0.999378, with no visible controlling divergence.

Salvage: after a compressed Remark 9.4, the candidate jumps directly to the bibliography and omits subsection (b), Proposition 9.5, Lemma 9.6, the X_N/Y_N construction, and the proof material. No page-local text beyond the inadequate remark was accepted.

English: translated only from the frozen French; “ordre de grandeur maximal” is rendered “maximal order of magnitude,” and all density, Hecke-operator, Frobenius-matrix, and asymptotic notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 24 / article page 23 / printed page 529.

Included matter: completion of the proof of Proposition 9.5; subsection (c); Proposition 9.7 and its proof; Remark 9.8; and Bibliographie [1]-[3]. High-risk checks covered $i(x) \sim cx/\log x$ and $\sum_{i \leq i(x)} \log p_i \sim cx$; the lower limsup bound; the normal order $2^{\log \log n}$; the density formula $1 - \prod_{p \in Y} (1 - (p-1)/p^2)$; divergence of $\sum_{p \in Y_N} 1/p$; the zero infinite product; $M(x) = o(x)$; and $M(x) = O(x/\log^\alpha x)$.

Source-faithful decisions: the proof's three intermediate asymptotics, inequality direction, finite-set density product, and the boundary between bibliography entries [3] and [4] were preserved. Only Bibliographie [1]-[3] belong to this physical page; entry [4] begins on physical page 25. Journal running matter was excluded.

Comparator: comparator page 23; same article page and geometry on direct review, with normalized text identity 1.000000.

Salvage: the candidate omits the completion of 9.5, all of subsection (c), Proposition 9.7, Remark 9.8, and the density/product argument. Its normalized bibliography cannot replace the authority wording or missing page body; accepted salvage remains zero.

English: standalone translation from the frozen French; bibliographic titles remain in their source languages while connective prose and page matter are translated, and all density, product, and asymptotic notation is preserved.

Assets: none. Unresolved readings: none.

Authority physical page 25 / article page 24 / printed page 530.

Included matter: Bibliographie [4]-[27]; “(Manuscrit reçu le 9 août 1974.)”; Pierre Deligne’s I.H.E.S. address; the connective “et”; Jean-Pierre Serre’s Collège de France address; and physical EOF.

High-risk authority readings checked at 400 dpi include [5] “La conjecture de Weil. I.”; [8] “3rd edit.”; [9] “(zw. Aufl.)” and “Göttingen”; [12] “à paraître aux Math. Ann.”; GL_2 in [13]; $\tau(n)$ in [18]; the Ω -result in [19]; “Über” and the nested parenthesis in [20]; accents and apostrophes in [21]-[23]; “Princeton Univ. Press.” in [24]; “Über” and “Funktionalgleichungen” in [26]; the separate “(Lezioni Fermiane). (Lecture Notes, ...)” construction in [27]; the date 9 août 1974; and both postal blocks.

Copy-matter decision: the running head and folio at the top, the terminal horizontal rule, and the journal footer “4e série — tome 7 — 1974 — n° 4” were excluded from the edition bodies. They remain visible in the authority and comparator evidence but are not work text.

Comparator: comparator page 24 is the same article page and geometry at lower raster. Its normalized text layer is byte-for-byte identical after normalization (identity 1.000000); direct rendered review disclosed no controlling divergence. The IAS authority remains controlling.

Salvage: the four byte-identical TeX witnesses contain bibliography [4]-[27] only in normalized form, with altered punctuation/diacritics and spelled-out tau/Omega, and omit the received-date line and both addresses. The seven derivative PDFs and two public-context members supply no acceptable page-local text. Accepted salvage remains zero.

English: standalone translation from the frozen French. Bibliographic titles stay in their source languages; conjunctions, page-range labels, the publication note in [12], the received-date line, and the address connective are rendered in English. Names, institutions, numbering, and postal data are preserved.

Assets: none. Unresolved readings: none.